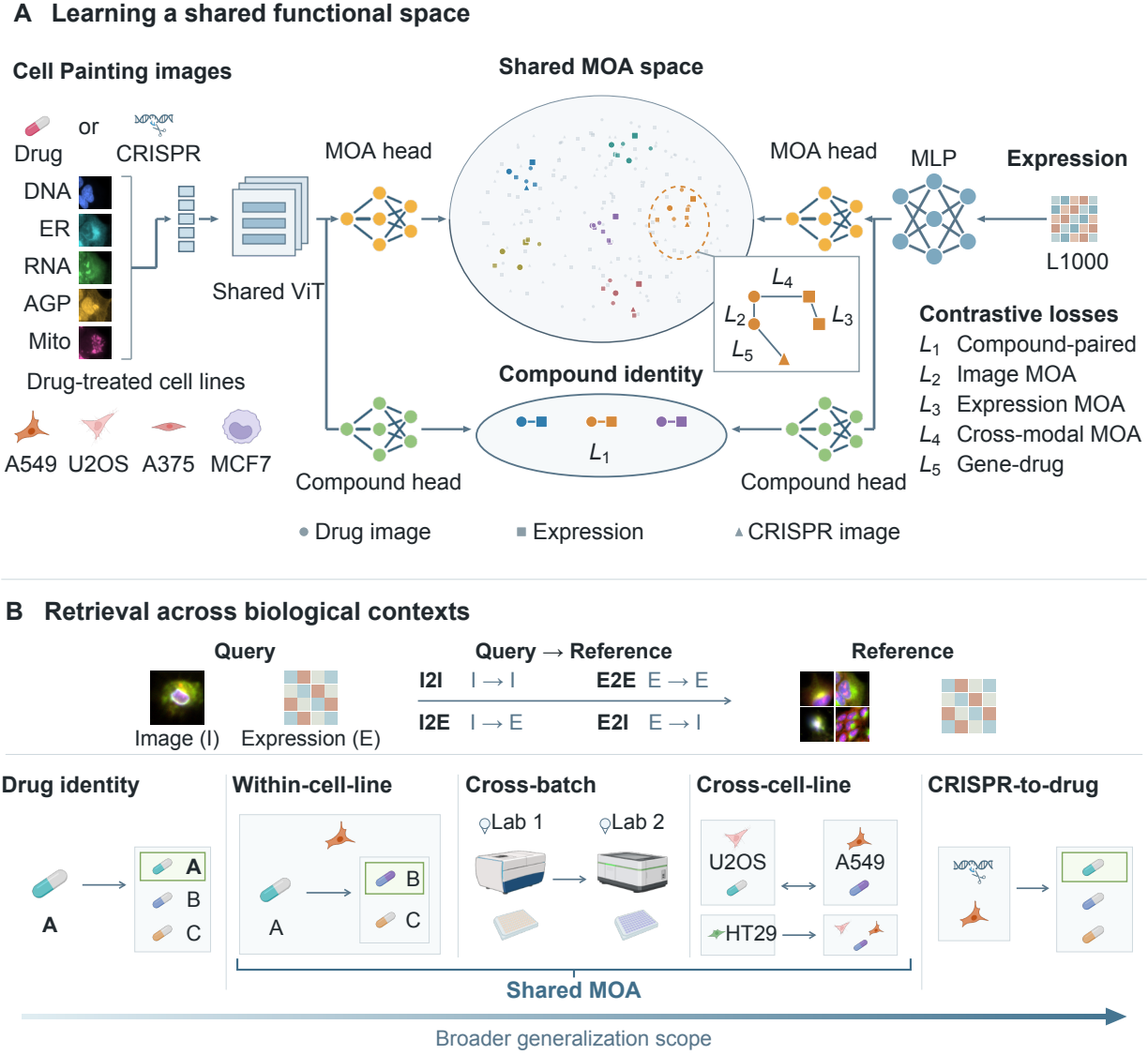

A Learning a shared functional space
Cell Painting images
Drug
or
CRISPR
DNA
ER
RNA
AGP
Mito
Drug-treated cell lines
A549
U2OS
A375
MCF7
Shared ViT
MOA head
Shared MOA space
MOA head
MLP
Expression
L1000
Contrastive losses
L_1 Compound-paired
L_2 Image MOA
L_3 Expression MOA
L_4 Cross-modal MOA
L_5 Gene-drug
L_4
L_2
L_3
L_5
Compound identity
L_1
Compound head
Compound head
Drug image
Expression
CRISPR image
B Retrieval across biological contexts
Query
Image (I)
Expression (E)
Query → Reference
I2I I → I
E2E E → E
I2E I → E
E2I E → I
Reference
Drug identity
A
A
B
C
Within-cell-line
A
B
C
Cross-batch
Lab 1
Lab 2
Cross-cell-line
U2OS
A549
HT29
CRISPR-to-drug
Shared MOA
Broader generalization scope